# Sanchit Ahuja

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sanchit-ahuja Google Scholar

## Research Interests

*AI Safety, Emergent Misalignment, Training Data and Model Behavior, Multilinguality, Culture*

## Education

**Northeastern University** — *Boston, MA, USA*
*PhD in Computer Science* — *Sept 2025 – present*

**Birla Institute of Technology and Science, Pilani** — *Pilani, India*
*Bachelor of Engineering in Electrical and Electronics* — *Aug 2017 – July 2021*

## Research Experience

**Pre-Doctoral Researcher** — *Bengaluru, India*
*Microsoft Research | Advisors: Sunayana Sitaram, Kalika Bali, Vishrav Chaudhary* — *Aug 2023 – July 2025*

- Showed that thinking in non-English languages EfficientXLang is token efficient without any loss in performance.
- Developed a large-scale synthetic dataset Updesh targeting 13 official Indian languages for instruction tuning.
- Developed MEGAVERSE, a comprehensive benchmark for multilingual and multimodal LLM evaluation.
- Created sPhinX, a framework for multilingual synthetic data generation and instruction tuning.
- Developed Scaling Laws for multilingual LLMs.
- Curated DOSA, a dataset of Indian social artifacts representing diverse subcultures.
- Published a comprehensive Contamination Report for multilingual benchmarks.

**Research Intern** — *Uppsala, Sweden*
*Uppsala University | Advisor: Meriem Beloucif* — *Sept 2021 – Nov 2023*

- Analyzed monolingual BERT models for semantic accuracy in token predictions
- Led the **2024 Semantic Relatedness task**, overseeing data annotation, baselines, and logistics.

**Research Intern** — *Hamburg, Germany*
*Universitat Hamburg, Germany | Advisors: Meriem Beloucif, Chris Biemann* — *Feb 2021 – July 2021*

- Applied RL to Neural MT to try and solve for the problem of Interactive Prediction Translation.

## Publications

**Specializing Without Forgetting: Analyzing Knowledge Preservation in Multilingual Model Adaptation**
***Sanchit Ahuja***, Terra Blevins
GenAI4World Workshop at COLM 2026 (**Oral Spotlight**) Code

**EfficientXLang: Towards Improving Token Efficiency Through Cross-Lingual Reasoning**
***Sanchit Ahuja‡***, Praneetha Vaddamanu‡, Barun Patra
SCALR workshop at COLM 2025 and EMNLP 2025 Findings Code

**Scaling Laws for Multilingual Language Models**
Yifei He, Alon Benhaim, Barun Patra, Praneetha Vaddamanu, ***Sanchit Ahuja***, Parul Chopra, et.al.
ACL 2025 Findings

**sPhinX: Sample efficient multilingual instruction fine-tuning through n-shot guided prompting**
***Sanchit Ahuja‡***, Kumar Tanmay‡, Hardik Hansrajbhai Chauhan, Barun Patra, Kriti Aggarwal, et.al
GEM² Workshop at ACL 2025

**Contamination Report for Multilingual Benchmarks**
***Sanchit Ahuja‡***, Varun Gumma‡, Sunayana Sitaram
EvalEval Workshop at NeurIPS 2024

**MEGAVERSE: Benchmarking Large Language Models Across Languages, Modalities, Models and Tasks**
***Sanchit Ahuja***, Divyanshu Aggarwal, Varun Gumma, Ishaan Watts, Ashutosh Sathe, et.al.
NAACL-HLT 2024 Code

**SemRel2024: A Collection of Semantic Textual Relatedness Datasets for 13 Languages**
Nedjma Ousidhoum, Shamsuddeen Muhammad, Mohamed Abdalla, Idris Abdulmumin, Ibrahim Ahmad, ***Sanchit Ahuja***, Alham Aji, Vladimir Araujo, Abinew Ayele, Pavan Baswani, Meriem Beloucif, Chris Biemann, et.al.
ACL 2024 Findings Code

**DOSA: A Dataset of Social Artifacts from Different Indian Geographical Subcultures**
Agrima Seth, ***Sanchit Ahuja***, Kalika Bali, Sunayana Sitaram
LREC-COLING 2024 Code

**HYPHEN: Hyperbolic Hawkes Attention For Text Streams**
Shivam Agarwal, Ramit Sawhney, ***Sanchit Ahuja***, Ritesh Soun, Sudheer Chava
ACL 2022 Code

## Selected Research Projects

**Analyzing Knowledge Preservation in Multilingual Model Adaptation** *Sept 2025 – June 2026*
- Localized catastrophic forgetting in multilingual continual pretraining via model interpolation; middle-layer reversion gave the largest comprehension recovery.
- Evaluated layer-informed mitigations (freezing, L2 regularization, reversion, souping); layer freezing exceeded base model comprehension.

**Improving Token Efficiency Through Cross-Lingual Reasoning** *Feb 2025 – June 2025*
- Investigated whether English is the most token-efficient language for reasoning in Language Reasoning Models across three open-source RLMs and seven typologically diverse languages.
- Demonstrated that reasoning in non-English languages reduces token usage while preserving accuracy, with gains persisting even after translating reasoning traces back to English.
- Established that improvement extent correlates with models multilingual strength, highlighting the importance of strong multilingual foundations for cross-lingual reasoning.

**Scaling Laws for Multilingual Language Models** *Sept 2024 – Jan 2025*
- Proposed a novel scaling law for decoder-only language models trained on multilingual data.
- Validated the hypothesis that CE loss for each language depends solely on its sampling ratio, independent of other languages in the mixture.
- Trained more than 100 models on 23 languages spanning 5 language families to validate the scaling law.

**Sample Efficient Multilingual Instruction Fine-Tuning Through N-shot Guided Prompting** *Feb 2024 – Oct 2024*
- Introduced a novel recipe for creating multilingual synthetic dataset by selectively translate instruction response pairs from English into 50 languages.
- Fine-tuned Mistral Base and Phi Base models using our dataset, achieving a 5%pt average performance improvement over the base models.
- Devised a strategy to incorporate N-shot examples in each fine-tuning sample that further boosted the performance by 9%pt and 4%pt respectively compared to vanilla fine-tuning.

**Benchmarking Large Language Models Across Languages, Modalities, Models and Tasks** *Aug 2023 – Dec 2023*

- Led the development of MEGAVERSE, a comprehensive benchmark for evaluating Large Language Models across languages, modalities, models, and tasks.
- Benchmarked a wide variety of models on 22 datasets covering 83 languages.
- Ran a data contamination study to establish that several models are likely to be contaminated with multilingual datasets.

## Engineering Experience

**Machine Learning Research Engineer** *Bengaluru, India*
*Skit.ai* *Sept 2021 – July 2023*

- Developed and deployed the first production model for noise detection in speech.
- Improved intent classification performance by **20%** with a contextual SLU model for critical clients.
- Led cross-team efforts to build benchmark datasets for intent, entity and ASR systems.
- Contributed to the open-source SLU library. Dialogy
- Improved SLU model confidence calibration, boosting macro-precision by **10%** for key clients. Blog

**Data Science Intern** *Bengaluru, India*
*Fi* *Apr 2021 – June 2021*

- Designed a Hierarchical Classification system for Transaction Categorization.

**Software Engineering Intern** *Bengaluru, India*
*ClearTax* *July 2020 – Dec 2020*

- Developed and enhanced the GST (Indirect Tax System of India) software for SMEs, CAs, and Enterprises.

**Machine Learning Intern** *Noida, India*
*Samsung* *May 2020 – July 2020*

- Optimized architectures for Single Image Super Resolution (SISR) with 2x and 4x upscaling.
- Implemented WDSR to achieve state-of-the-art PSNR of 30 on the Div2K dataset.

## Teaching and Volunteering Roles

**Natural Language Processing Reading Group** *Nov 2023 – July 2025*
*Microsoft Research, India*

**Teaching Assistant** *Jan 2021 – May 2021*
*BITS Pilani, India*